

TASK 2

Case Study Analysis: Moderna

mRNA biotechnology, vaccines, therapeutics and the business of platform innovation

PREPARED BY

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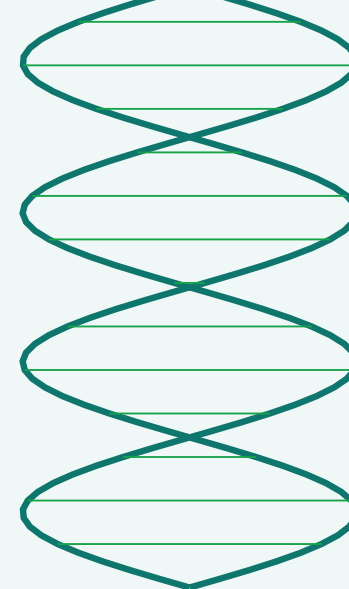
Domain: Biotechnology

Focus: Biotechnology innovation, applications, challenges and market impact

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mRNA PLATFORM



Design → deliver → express protein → transient message

1. Company Overview

Why Moderna is a useful biotechnology case study

Moderna, Inc.

- A biotechnology company built around messenger RNA (mRNA) medicines.
- Its platform combines mRNA science, delivery technologies and advanced manufacturing.
- The company develops vaccines and therapeutics across infectious diseases, cancer and rare diseases.
- Its strategy is to reuse a common technological platform across multiple products and disease areas.

Platform model



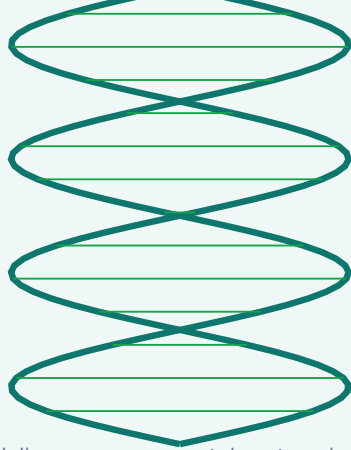
Design optimized mRNA the message in the code. Produce consistently at scale. Test, approve and commercialize

Source: Moderna, 'Moderna's mRNA Platform' and 'Therapeutic Areas'.

2. Core Biotechnology Innovation

How Moderna's mRNA platform works

mRNA PLATFORM



Design → deliver → express protein → transient message

The basic biological concept

- Scientists identify a protein that could help prevent or treat a disease.
- They design an mRNA sequence that carries instructions for making that protein.
- The mRNA is delivered into cells, often using lipid-based delivery systems.
- Cells temporarily use the mRNA to make the target protein.
- The mRNA is naturally broken down after it has served its function; it does not permanently alter DNA.

Key biotechnology principle: deliver transient genetic instructions rather than permanently changing the genome.

3. Products and Pipeline

From COVID-19 vaccines to a broader mRNA medicine portfolio

Commercial products highlighted by Moderna in 2026

- SPIKEVAX®: COVID-19 vaccine.
- mRESVIA®: RSV vaccine.
- mNEXSPIKE®: next-generation COVID-19 vaccine.

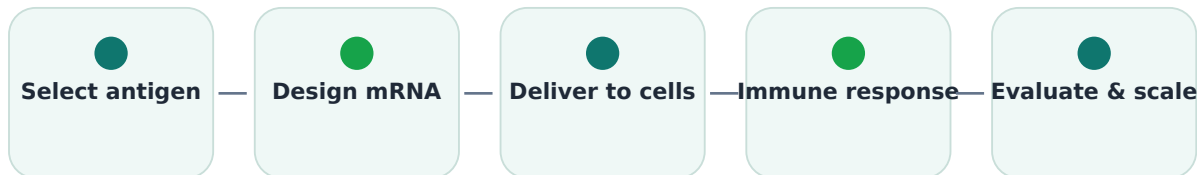
Pipeline direction

- Respiratory vaccines
- Cancer / individualized neoantigen therapies
- Rare-disease therapeutics
- Additional infectious-disease vaccines

Pipeline status is dynamic; the cited Moderna pipeline page is dated May 28, 2026.

4. Major Application: Vaccines

Why mRNA became a strategically important vaccine technology



- mRNA allows researchers to encode instructions for a selected antigen without needing to manufacture the antigen itself as the starting biological material.
- The platform can support rapid design-test cycles, which is valuable when pathogens change or new public-health threats emerge.
- Moderna has expanded its respiratory portfolio beyond its original COVID-19 vaccine to include RSV and next-generation COVID-19 products.

Biotechnology significance

The same design and manufacturing principles can be adapted to multiple antigens, making the platform reusable.

5. Therapeutic Applications

Beyond vaccines: cancer and rare-disease medicine

Cancer

- Individualized neoantigen approaches aim to train the immune system to recognize patient-specific tumor targets.
- mRNA can encode multiple antigens in one therapeutic design.
- The goal is personalized immuno-oncology rather than a single universal cancer vaccine.

Rare diseases

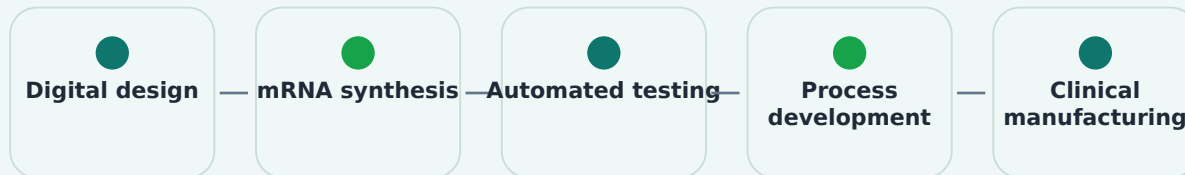
- mRNA can potentially provide cells with instructions to make a missing or defective protein.
- This creates a route toward treating disorders caused by inadequate protein production.
- Delivery to the right tissues and durable clinical benefit remain major development challenges.

Platform advantage

- One core technology can be adapted to different proteins and diseases.
- Design, delivery and manufacturing knowledge can potentially be reused across programs.
- This is the central business logic of platform biotechnology.

6. Manufacturing and Digital Biotechnology

How the company turns biological design into repeatable development



- Moderna describes an mRNA Design Studio that uses computational tools to design and optimize mRNA sequences.
- Its research engine combines digital drug-design capabilities with automated production to support parallel development.
- High-throughput workflows can shorten the time needed to move from a protein concept to candidate mRNA constructs.
- This integration of biology, software, automation and manufacturing is a defining feature of Moderna's biotechnology model.

Source: Moderna, 'Moderna's mRNA Platform' and 2026 Science Day materials.

7. Challenges

Scientific, manufacturing, regulatory and commercial constraints

Delivery

mRNA is fragile and must reach the correct cells efficiently; tissue targeting remains a major research problem.

Durability

Transient expression is useful for many applications, but some diseases may require repeated or sustained treatment.

Manufacturing

Quality, scale, consistency and cost must be controlled for every product and formulation.

Clinical uncertainty

Promising laboratory results do not guarantee efficacy or safety in diverse patient populations.

Market dynamics

Seasonal vaccines face competition, changing demand and pricing pressure.

Access & equity

Advanced biologics can require specialized infrastructure and may be difficult to distribute equitably.

8. Market Impact

How Moderna changed the biotechnology and vaccine landscape

Commercial evidence

- Moderna reported 2025 projected revenue of \$1.6–\$2.0 billion in its shareholder letter, reflecting its commercial vaccine franchise.
- The company reported that mNEXSPIKE represented about 24% of U.S. retail COVID vaccinations in 2025 and nearly one-third among adults aged 65+.
- The company is using its commercial vaccine base to support expansion into oncology and rare diseases.

Broader industry impact

- Validated mRNA as a commercial vaccine platform.
- Accelerated investment in RNA medicines and delivery technologies.
- Demonstrated the value of platform-based biotechnology, where one technical foundation supports multiple products.
- Raised the importance of scalable lipid delivery, automated manufacturing and digital biological design.

Market figures are company-reported and should be interpreted alongside the wider seasonal vaccine market.

9. SWOT Analysis

A concise strategic view of Moderna as a biotechnology platform company

STRENGTHS

- Deep mRNA expertise
- Integrated design, delivery and manufacturing capabilities
- Multiple approved products and clinical programs

WEAKNESSES

- Revenue concentration in respiratory vaccines
- High R&D and manufacturing costs
- Dependence on successful clinical translation

OPPORTUNITIES

- Cancer immunotherapy
- Rare-disease therapeutics
- New infectious-disease vaccines
- AI and automation for faster discovery

THREATS

- Strong competition
- Changing vaccine demand
- Regulatory and safety uncertainty
- Pricing and access pressure

10. Conclusion & References

Key takeaways for biotechnology students

Conclusion

- Moderna demonstrates how a core molecular technology can become a scalable biotechnology platform.
- Its mRNA approach supports vaccines and emerging therapeutic applications
- The strongest long-term opportunity is platform reuse across diseases, supported by delivery science, manufacturing and digital tools.
- The main barriers are not only biological. Cost, manufacturing, regulation, market demand and equitable access also determine impact.

Selected references

Moderna. (2026). Moderna's mRNA platform.
Moderna. (2026). mRNA pipeline (as of May 28, 2026).
Moderna. (2026). Building Moderna's next decade of growth.
Moderna. (2025). 2025 shareholder letter.
Moderna. (2026). Therapeutic areas & modalities.
Moderna. (2026). Science of mRNA.

Full source links:
modernatx.com/power-of-mrna
modernatx.com/research/product-pipeline
investors.modernatx.com